



Silicone low surface energy antifouling coating modified by zwitterionic side chains with strong substrate adhesion

Dejin Zhang, Susu Zhao, Zhihao Rong, Kaiming Zhang, Chuanhui Gao, Yumin Wu, Yuetao Liu*

College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, China

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ABSTRACT

Silicone fouling release coatings were receiving increasing concern due to of their eco-friendly properties. However, their poor fouling resistance capability and mechanical properties limited the applications. Herein, we reported a type of silicone low surface energy coating with zwitterionic side chains and the carbamates group was introduced into the Si-O main chain. The carbamates groups and zwitterions provided strong substrate adhesion up to 2.0 MPa through hydrogen bond and ion interaction which also endowed the good mechanical properties higher than 6.0 MPa. The coating had a low surface energy less than $30 \text{ mJ}\cdot\text{m}^{-2}$, and exhibited good fouling release properties in the range of 0.3–0.5 MPa. Also, zwitterions gave the coating excellent fouling resistance properties. The bioassays showed that the anti-bacterial efficiency of PDMS-zPDEM-14 coating against *Pseudomonas* sp. was 94.86%, the coating could inhibit the formation of biofilm (the coverage area was only 13.64%) and effectively prevent the adhesion of diatoms (the density of adhered diatoms was $107 \text{ cells}/\text{mm}^2$). This work had promising applications in low surface energy marine antifouling coating and interfaces of biomaterials.

1. Introduction

Marine biofouling was formed by the accumulation of bacteria, diatoms and other organisms in seawater on solid surfaces [1,2]. Marine fouling organisms were becoming the global concern because of the negative impact on marine vessels, facilities and ecosystems [3–5]. Currently, the main antifouling strategy was to use paints containing pesticides to kill or inhibit the attachment of fouled organisms [6,7]. However, the biocide-containing antifouling coatings were toxic, affecting other harmless marine organisms and endangering the marine ecological balance [8,9]. Therefore, the antifouling coatings containing pesticides had gradually been banned. In recent years, a variety of eco-friendly anti-fouling coatings had been reported, including fouling release coatings [10–13], degradable polymers [14,15], anti-protein polymers [16,17], amphiphilic polymers [18–22], zwitterionic polymers [23–28] and biomimetic antifouling materials [29,30].

Silicones had low surface energy and modulus, which could weaken the adhesion of fouling organisms, so the fouling organisms could be easily removed under the action of external forces, and its anti-fouling effect did not depend on the release of antifoulants [31,32].

Therefore, silicones were receiving more and more attention in marine antifouling coatings. However, the low antifouling ability under static conditions, weak substrate adhesion, and poor mechanical properties which were easily destroyed by external forces, limited their antifouling applications. Amphiphilic polymers containing both hydrophobic and hydrophilic chain segments utilized the protein-resistant properties to stop biological contamination and were effective in inhibiting the adhesion of fouling organisms [33]. Therefore, amphiphilic modification was an effective way to improve the static antifouling ability of silicones.

Polyethylene glycol (PEG) was a commonly used non-toxic material that resists protein absorption. It could bond with water molecules by hydrogen bond to inhibit protein binding [34–37]. However, PEG would self-oxidize relatively quickly in the presence of oxygen and transition metal ions [38]. The stability was not good enough and it could not be used as antifouling coatings for a long time [39–41]. Zwitterionic polymers were another kind of hydrophilic antifouling materials. Compared with PEG, they had a positively and negatively charged groups on the same repeating unit, and their hydration was stronger, which could bind more strongly to water through electrostatic induction

* Corresponding author.

E-mail address: yuetaoliu@qust.edu.cn (Y. Liu).

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[42,43], where betainephosphate, sulfobetaine and carboxybetaine were zwitterionic candidates [44–47]. However, the introduction of zwitterion into the silicone polymer as an antifouling group would cause the reduction of mechanical strength [22].

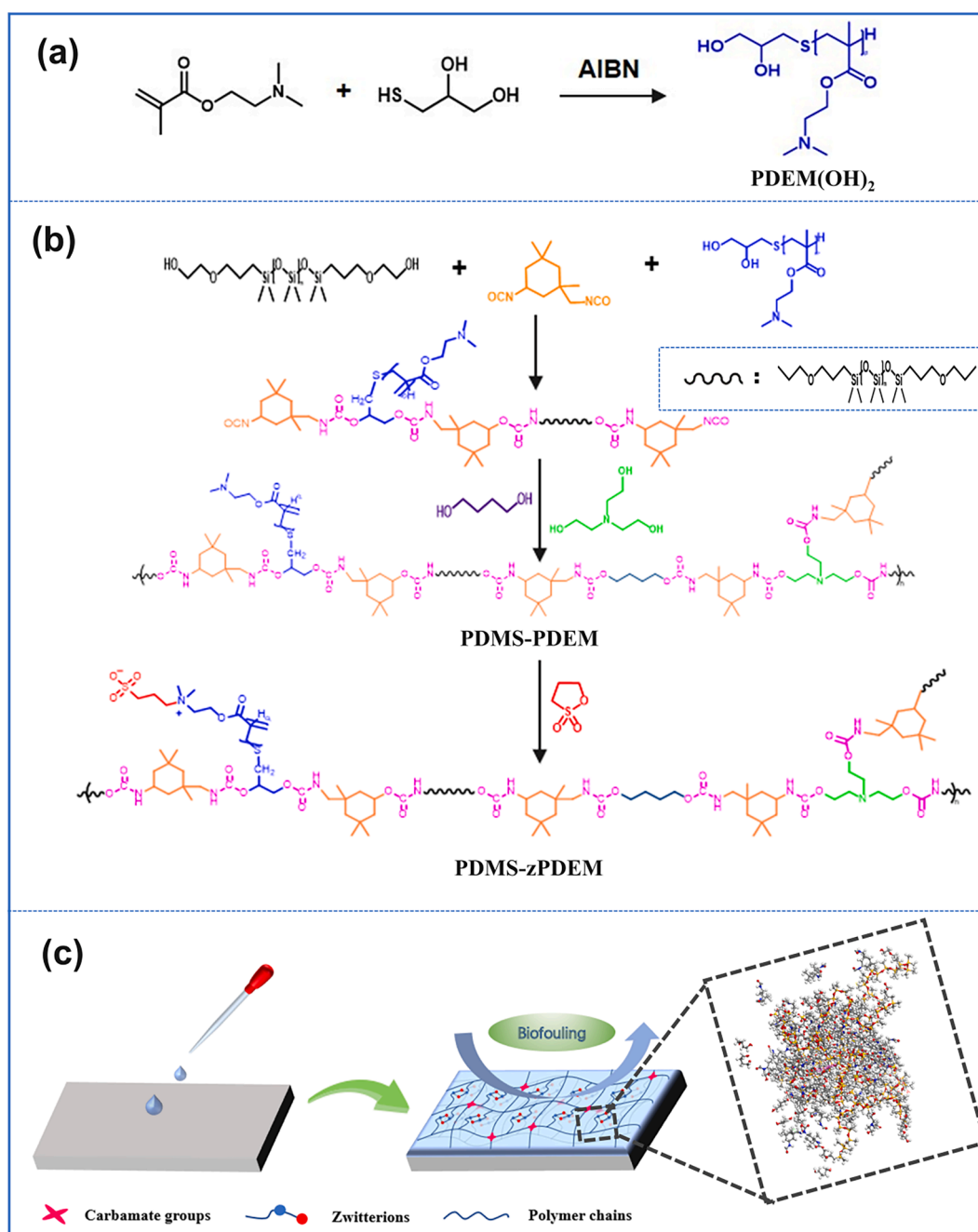
In this study, the hydroxylated zwitterionic precursors were synthesized by free radical polymerization with 2-(dimethylamino) ethyl methacrylate (DEM) and 3-mercapto-1,2-propanediol (TPG), and introduced into the silicone side chains by the addition polymerization of hydroxyl and isocyanate groups. The zwitterions were formed under the action of 1,3-propanesultone (1,3-PS). The nature of Si-O main chains provided the coating with low surface energy. The large amount of hydrogen and ionic bonds provided the strong substrate adhesion. The zwitterionic side chains formed a hydration layer on the surface of the coating through electrostatic interaction with water molecules, thus improving the antifouling performance. The mechanical properties,

adhesion, wettability, fouling release ability and stability of the coatings and the resistance to marine bacteria, biofilms and diatoms were also investigated.

2. Materials and methods

2.1. Materials.

Hydroxyl terminated polydimethylsiloxane (PDMS(OH)₂, M_n = 2000 g/mol) was from Jian de Polymer New Materials Co., Ltd, and distillation under reduced pressure at 120 °C for 4 h before use. 2-(dimethylamino) ethyl methacrylate (DEM), 3-mercapto-1,2-propanediol (TPG), 3-(isocyanatomethyl)-3,5,5-trimethylcyclohexyl isocyanate (IPDI), triethanolamine (TEOA), dibutyltin dilaurate (DBTDL), 1,3-propanesultone (1,3-PS) were purchased from Aladdin and used as



Scheme 1. Synthesis process of PDEM(OH)₂ (a). Synthesis process of PDMS-zPDEM-x (b). Schematic illustration of PDMS-zPDEM-x coatings preparation (c).

received. 1,4-butane diol (1,4-BD) from Aldrich was distilled under reduced pressure for 4 h before use. 2,2-azobisisobutyronitrile (AIBN) was purchased from Macklin and used after recrystallization twice from methanol. N-hexane and tetrahydrofuran (THF) were provided by Sinopharm and THF was dehydrated with 4A molecular sieve for 24 h before use. PDMS elastomer (Sylgard 184) was purchased from Dow Corning, it was composed of two components of A and B. when used, the two components were mixed in a mass ratio of 10:1. Wherein, component A was mainly composed of α , ω -vinyl terminated polydimethylsiloxane, vinyl resin, polydimethylsiloxane polymethylhydrosiloxane copolymer, and component B was mainly a mixture containing platinum catalyst. Artificial seawater (ASW) was prepared according to standard ASTM D1141-98 (2013).

2.2. Synthesis of PDEM(OH)₂.

As shown in Scheme 1a, PDEM(OH)₂ was synthesized by radical polymerization. TPG was used as chain terminator and diol group was introduced. In a typical experiment, DEM (50 g, 318 mmol), TPG (3.44 g, 31.8 mmol), AIBN (0.075 g, 0.15 wt% based on monomers), and THF (240 mL) were added to a three-neck flask connected to the condensing unit in nitrogen atmosphere and the reaction was stirred at 70 °C for 12 h. The product was precipitated with excess n-hexane and dried in vacuum at 50 °C for 24 h to obtain the pale yellow products. Polydispersity index ($\overline{DM}$) and M_n of PDEM(OH)₂ were respectively 1.55 and 1750 g/mol, which were measured by gel permeation chromatography (GPC).

2.3. Preparation of PDMS-zPDEM-x coatings.

PDMS-zPDEM-x was synthesized by an addition polymerization using THF as solvent (Scheme 1b). The molar ratio of NCO:OH was 1.1. IPDI and PDMS(OH)₂ were added to a three-neck flask connected to a condensing unit under nitrogen atmosphere and the reaction was stirred at 70 °C for 12 h, then PDEM(OH)₂ was added and the reaction was continued for 1 h. After that, 1,4-BD and DBTDL were added and the reaction carried out at 80 °C for 3 h. Then TEOA was added and the mixture was continued for 1 h. Finally, the system was cooled to room temperature and 1,3-PS was added to react for 24 h to obtain PDMS-zPDEM-x. PDMS-PDEM-14 was synthesized in the same way as PDMS-zPDEM-x, except that 1,3-PS was absent. The detailed formulation was shown in Table 1. The 2 mL obtained polymer solution (30 wt%) was coated on the epoxy plates (50 × 50 mm²) by drop casting and solidified at room temperature to obtain the coating (Scheme 1c).

2.4. Characterizations

Proton nuclear magnetic resonance spectroscopy (¹H NMR). The Bruker 400 M spectrometer was used and CDCl₃ was selected as the solvent for testing at 25 °C. Fourier Transform Infrared Spectroscopy (FTIR). The Nicolet IS10 Fourier was used and each measurement range was 4000–400 cm⁻¹. Gel permeation chromatography (GPC). The molecular weight (M_n) and the polydispersity ($\overline{DM}$) were measured by GPC with PL-GPC50. The test was performed at 40 °C with TSKgel SuperMutiporeHZ-H column. THF was used as eluent, and the flow rate

was 0.35 mL·min⁻¹. Monodisperse polystyrene was the standard. X-ray Photoelectron Spectroscopy (XPS). Thermo Scientific K-Alpha electron spectrometer was used. The excitation source was Al K α rays ($h\nu = 1486.6$ eV), the pressure of the sample chamber was less than 2.0×10^{-7} mbar, the operating voltage was 12 kV, the filament current was 6 mA, and the energy standard was charge-corrected with C_{1s} = 284.80 eV binding energy. Thermogravimetric analysis (TGA). The tests were performed by a simultaneous thermal analyzer STA449F3 (NETZSCH) at a nitrogen flow rate of 50 mL/min. The sample mass was approximately 10 mg and each sample was scanned between 298 and 1073 K with a heating rate of 10 K/min.

Tensile tests. The tests were performed by a CMT4204 universal testing machine and the tensile rate was 50 mm/min. The samples were dumbbell type (length × width: 50 × 4 mm³). Each sample was measured five times at 25 °C.

Adhesion tests. The adhesion between coating and epoxy substrate was measured by pull-off tests using CMT4204 universal testing machine and the tensile rate was 5 mm/min. The adhesion measured by pulling tests referred to the uniform and vertical force applied above the cemented surface at the specified speed, which was used to determine the required force when the adhesion between the coating and the substrate was destroyed.

Contact angle and surface energy tests. The tests were performed at 25 °C by a JC2000D1 contact angle measuring instrument. 2 μ L deionized water or 1 μ L diiodomethane was dropped onto the coating surface using sessile method. Each sample was measured five times and averaged. The surface energy was calculated by the Owens-WendtRabel-Kaelble method based on the measured contact angles of water and diiodomethane [48].

Pseudobarnacle removal tests. The tests were conducted following the standard ASTM D5618 (2011). The aluminum ingots (height 10 mm, diameter 10 mm) and the coating surface were fixed via an epoxy adhesive. One end of the force gauge (HP-500 N) was attached to the ingot and the force was applied in the horizontal direction at a constant speed, and the numeric value of the gauge was recorded when the aluminum ingot is separated from the coating surface.

Mass loss tests. 10 g of 30 %wt polymer solution was prepared and formed into a film in a PTFE mold, cut into an area of 10 × 10 mm² and the mass (W_1) was recorded after complete curing. Afterward the sample was immersed in artificial seawater (ASW) at 25 °C. After 21 days, the sample was removed and the surface was gently rinsed with deionized water, and then the sample was placed in a vacuum drying oven at 70 °C. After three days of drying, the sample was removed and its mass (W_2) was recorded. The mass loss rate (WL) was calculated by equation (1):

$$WL = \left(1 - \frac{W_2}{W_1}\right) \times 100\% \quad (1)$$

Three parallel samples were taken for each sample to calculate the average value.

Antibacterial tests. Pseudomonas sp. was selected to evaluate the antibacterial properties of the coatings. Firstly, the sample solution with the concentration of 30 %wt was formed into a film in the PTFE mold. The samples were sterilized with a 20 W, 253.7 nm UV lamp for 30 min. The samples were loaded into petri dishes and 10 mL diluted bacterial solution (10⁸ CFU/mL) was added, and then the petri dishes were placed in a constant temperature incubator and incubated for 12 h at 30 °C. After culture, the medium was serially diluted 10³ times with LB and 10 μ L of the diluted liquid medium sample was evenly scraped onto LB solid medium and incubated for 48 h at 30 °C. The number of adherent colonies of different types of polymers was determined by plate counting method. The number of colonies grown on solid medium was recorded as N_1 . The number of colonies in the control group was recorded as N_2 . The antibacterial rate (AE) of PDMS-zPDEM-x were calculated by Eq. (2):

Table 1
The experimental formulation of different coatings, unit (g).

Sample	PDMS (OH) ₂	IPDI	PDEM (OH) ₂	1,4-BD	TEOA	1,3-PS
PDMS-zPDEM-0	10	3.5	0	0.77	0.15	0
PDMS-PDEM-14	10	3.5	1.31	0.7	0.15	0
PDMS-zPDEM-5	10	3.5	0.43	0.74	0.15	0.36
PDMS-zPDEM-10	10	3.5	0.88	0.72	0.15	0.73
PDMS-zPDEM-14	10	3.5	1.31	0.7	0.15	1.08

$$AE = \left(1 - \frac{N_1}{N_2}\right) \times 100\% \quad (2)$$

Anti-biofilm tests. The biofilm formation was promoted by prolonging the culture time of *Pseudomonas* sp. 1 mL of bacterial suspension (10^8 CFU/mL) was introduced into a 24-well plate containing the sample. After the bacterial was incubated at 30 °C for 72 h, the bacterial solution was removed with a pipette and carefully rinsed 3 times with ASW to remove the unattached biofilm. Then the biofilm was fixed on the coating with methanol, stained with crystalline violet, excess

crystalline violet was rinsed with ASW, the biofilm was photographed with a digital camera, and the area covered by the biofilm was calculated with ImageJ software. Finally, the crystalline violet was extracted with acetic acid and the optical density of the biofilm was measured at $\lambda = 589$ nm.

Anti-diatom tests. Sterilized seawater and diatoms (*Pleurosigma* sp.) were added to a conical flask in a ratio of 3:1, and the appropriate amount of nutrient solution was added to the conical flask, then the flask was sealed with kraft paper and twine, and finally the diatom culture

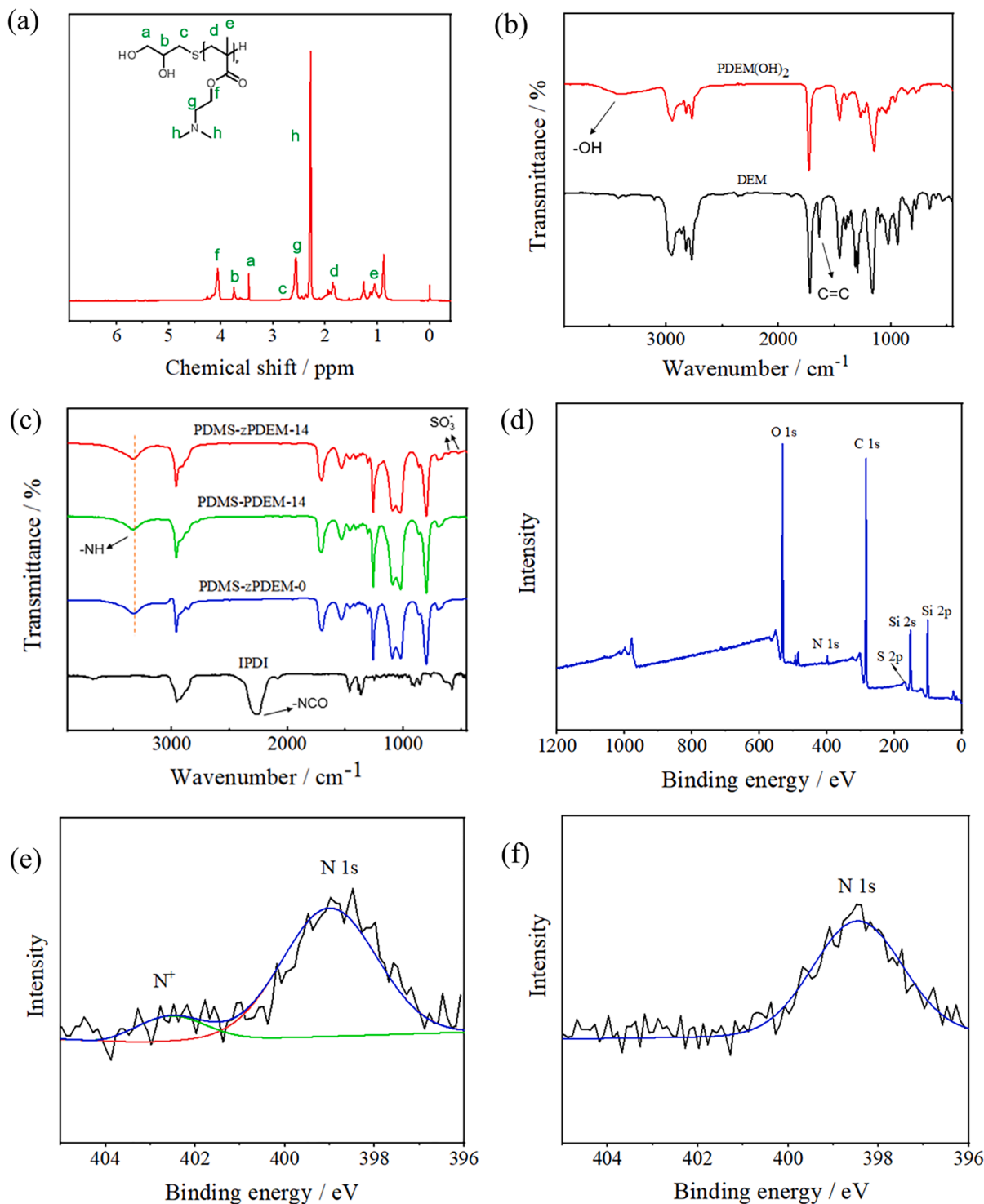


Fig. 1. ^1H NMR of PDEM(OH)₂ (a). FTIR spectrum of DEM and PDEM(OH)₂ (b). FTIR spectrum of IPDI, PDMS-zPDEM-0, PDMS-PDEM-14 and PDMS-zPDEM-14 (c). XPS spectra of PDMS-zPDEM-14 (d). XPS spectra of N_{1s} atoms of PDMS-zPDEM-14 (e). XPS spectra of N_{1s} atoms of PDMS-PDEM-14 (f).

was incubated in an artificial climate chamber at 25 °C for 72 h at a 14 h:10 h light-to-dark ratio. After the culture was completed, the diatom suspension was diluted to 1×10^5 cell/mm² with sterilized seawater, and the samples were placed in 24-well plates and the diluted diatom suspension was added to them and incubated for 24 h. After incubation, the samples were carefully rinsed with sterilized seawater to remove the unattached diatoms, observed with an inverted microscope and counted in five fields of view. Three parallel samples were used for each coating to obtain an average.

3. Results and discussion

3.1. Structural analysis

The structure of PDEM (OH)₂ was characterized by ¹H NMR and FTIR. The PDMS-zPDEM-x coating was unable to dissolve by suitable deuterated reagent, so the structure of the target products was confirmed by FTIR and XPS. Fig. 1a showed the ¹H NMR spectrum of PDEM(OH)₂. The proton peaks in the spectrum corresponded to PDEM (OH)₂ one by one, where peaks a, b and c were attributed to the TPG structure, peaks d, e, f, g and h were attributed to the DEM structure. Fig. 1b showed the FTIR spectra of DEM, PDEM(OH)₂ and IPDI. The C=C peak at 1640 cm⁻¹ had disappeared and the OH peak at 3450 cm⁻¹ in PDEM(OH)₂ had appeared compared to the monomeric DEM, further

demonstrating that PDEM(OH)₂ had been successfully synthesized. As shown in Fig. 1c, for PDMS-zPDEM-0, PDMS-PDEM-14 and PDMS-zPDEM-14, the appearance of the characteristic peak of NHCOO at 3307 cm⁻¹, the disappearance of the characteristic peak of NCO at 2260 cm⁻¹ and the OH peak at 3450 cm⁻¹ indicated the successful synthesis of the target polymer and PDEM had been successfully introduced. For PDMS-zPDEM-14, the appearance of SO₃⁻ characteristic absorption peaks at 605 cm⁻¹ and 524 cm⁻¹ indicated that the side chains were successfully sulfonated. Fig. 1d was the XPS spectrum of PDMS-zPDEM-14. The peak of S_{2p} was detected at 168 eV, and only the zwitterionic chain segments contained S element, indicating that they were distributed on the coating surface. Fig. 1e and f showed the XPS spectra of N_{1s} atoms of PDMS-zPDEM-14 and PDMS-PDEM-14, respectively. 400 eV corresponded to the peak of N element in PDEM or polyurethane. For PDMS-zPDEM-14, in addition to 400 eV, a displacement peak had appeared at 402.5 eV, which was the peak of N element on the sulfonamide group. The results of XPS also showed that zwitterionic polymers were introduced into the silicone side chains.

3.2. Thermal stability

Fig. 2a showed the TGA curves of PDMS-zPDEM-0 and PDMS-zPDEM-14. PDMS-zPDEM-14 had a mass loss at 100 °C, which might be caused by the adsorption of water by zwitterionic side chains

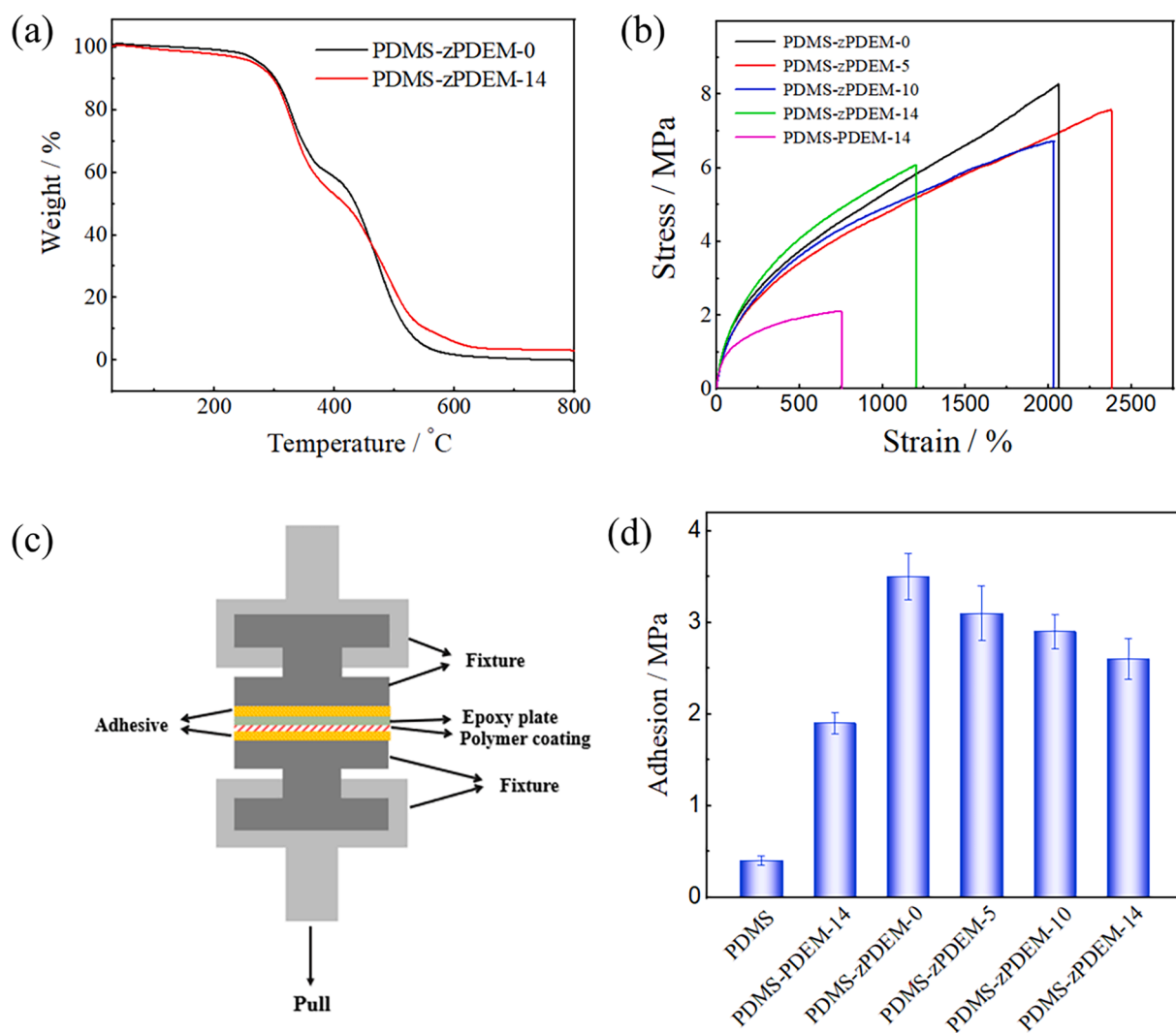


Fig. 2. TGA curves of PDMS-zPDEM-0 and PDMS-zPDEM-14 (a). Stress-strain curves of PDMS-zPDEM-x (b). Schematic diagram of pull-off tests for coatings adhesion measurement (c). Adhesion strength of PDMS-zPDEM-x coatings on epoxy plates (d).

(zPDEM). The thermal decomposition temperatures of PDMS-zPDEM-0 and PDMS-zPDEM-14 were both above 260 °C, indicating that they both had good thermal stability. The 5% thermal weight loss temperature of PDMS-zPDEM-0 was 276 °C higher than PDMS-zPDEM-14 indicating that the introduction of zPDEM could reduce the thermal stability. In addition, the ash content of PDMS-zPDEM-14 increased to

3%, which was the reason for the introduction of very stable S element through the PS unit. These phenomena provided another strong evidence for the successful introduction of zwitterionic polymers into silicone side chains.

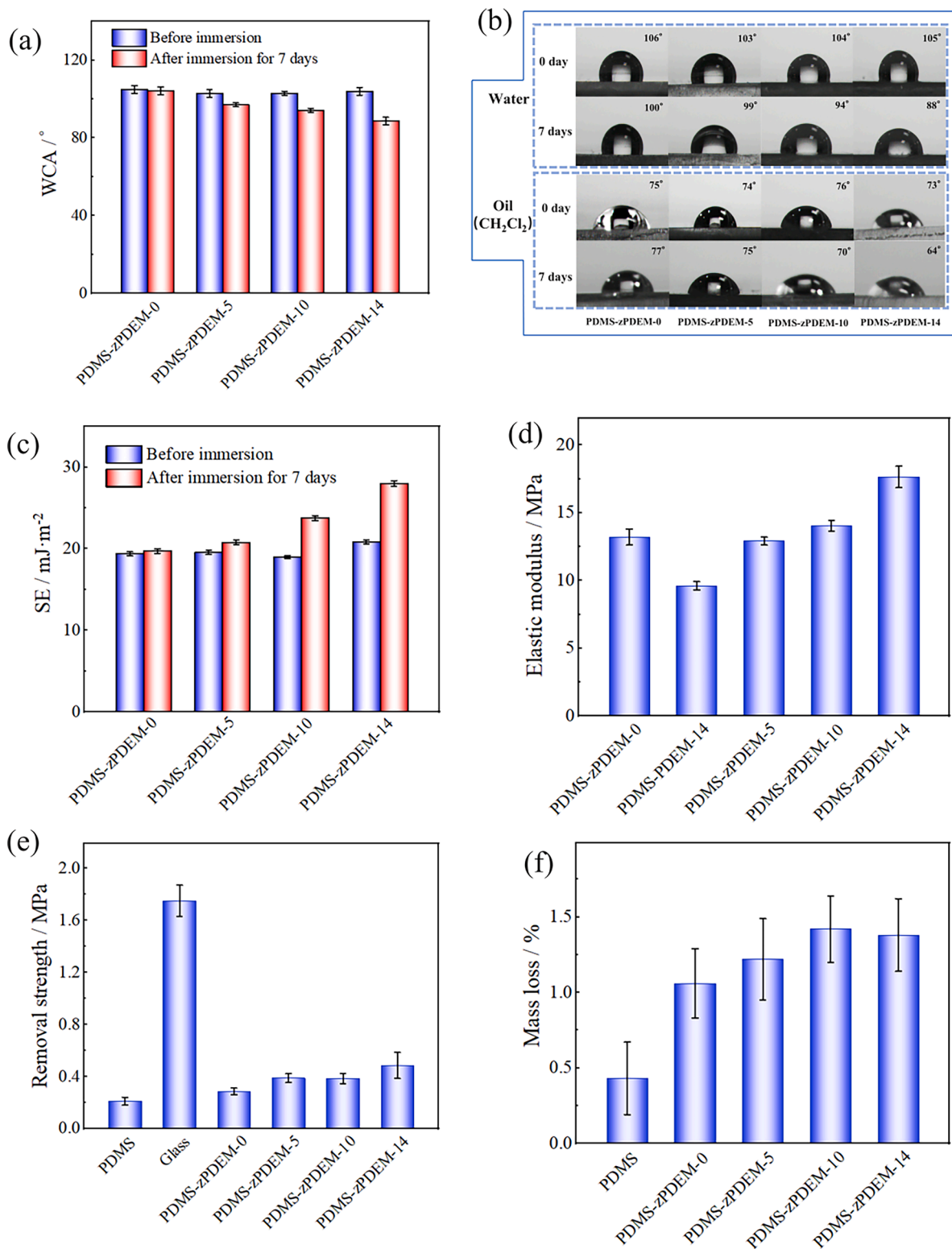


Fig. 3. Static water CA (a), water and oil (diiodomethane) CA images (b) and SE (c) before and after immersion in ASW for 7 days. Elastic modulus of PDMS-zPDEM-x (d). Removal strength of pseudobarnacles of PDMS-zPDEM-x (e). Mass loss of PDMS-zPDEM-x after immersing in ASW for 21 days (f).

3.3. Mechanical properties

Excellent mechanical properties could prevent physical damage of the coating during service, so the stress–strain curves of silicone coating were conducted. Fig. 2b showed the mechanical properties of coatings. PDMS-zPDEM-0 had good strength and flexibility, the stress was 8.28 MPa and the strain was 2061%. Compared with PDMS-zPDEM-0, PDMS-PDEM-14 introduced macromolecular side chains and the tensile strength was decreased to 2.11 MPa and elongation at break to 754%, indicating that the introduction of macromolecular side chains destroyed the hydrogen bond interaction and reduced the mechanical properties of the materials. However, PDMS-zPDEM-14 produced zwitterions, its tensile strength (6.07 MPa) and elongation at break (1202%) were improved because of the electrostatic interaction between zwitterions which made up for the decline of mechanical properties. In general, with the increase of zPDEM, the tensile strength of the coating decreased from 7.59 MPa for PDMS-zPDEM-5 to 6.07 MPa for PDMS-zPDEM-14, and the elongation at break decreased from 2379% for PDMS-zPDEM-5 to 1202% for PDMS-zPDEM-14. Despite the introduction of zwitterion segments reduced the mechanical properties of the material, the material still has high mechanical properties. Also, the adhesion of the coating epoxy plates was tested by pull-off method.

Fig. 2c and d showed the test device and the adhesion strength epoxy plates, PDMS-zPDEM-0 had the highest adhesion strength, and the adhesion strength of PDMS-zPDEM-x decreased gradually with the increase of zPDEM, because it decreased the hydrogen bond density between coating and substrate polar groups. In addition, the adhesion of PDMS-zPDEM-14 (about 3.5 MPa) was higher than that of PDMS-PDEM-14 (about 1.9 MPa), because the ionic interaction between zwitterions and substrate improved the adhesion [49,50]. Overall, the adhesion strength of all PDMS-zPDEM-x coatings was much higher than that of 1.0 MPa (2.6–3.5 MPa), indicating that it was sufficient for marine antifouling coatings applications [51]. On the contrary, the adhesion strength of PDMS as a control sample was weak (about 0.3 MPa) as its non-polar structure made it easy to separate from the substrate [52].

3.4. Wettability, fouling release performance and stability

As shown in Fig. 3a, b and c, PDMS-zPDEM-x had water contact angle (WCA) higher than 100° and low surface energy (SE) from 19 mJ·m⁻² to 21 mJ·m⁻² before immersion in ASW. With the increase of zPDEM, WCA and SE changed slightly, because a large number of Si-O chain segments that had lower SE in the coating preferentially covered the outermost surface under air conditions [53,54]. After 7 days of immersion in ASW, WCA significantly decreased and the SE increased, as the zPDEM exhibited very low SE when the coating was immersed in ASW leading to the surface reconstruction and migration of hydrophilic chain segments to the surface [55–57]. Although the introduction of zPDEM changed the surface wettability of the coatings, it still had a low surface energy after immersion in ASW.

Removal strength of pseudobarnacles was an important indicator to evaluate fouling release performance of coatings. The lower the removal strength (RS), the better the fouling release ability of the coatings. Studies showed that the fouling release performance of the coating was related to the surface energy and elastic modulus [58–60]. As shown in Fig. 3d, the elastic modulus gradually increased from 12.91 MPa to 17.64 MPa with the increase of the zPDEM. The elastic modulus of PDMS-zPDEM-14 was 17.64 MPa much higher than that of PDMS-PDEM-14 (9.62 MPa), because the cross-linking effect by the zwitterions limited the movement of the chain segments. The fouling release ability of the coatings was evaluated using the removal strength of pseudobarnacles tests. As shown in Fig. 3e, the PDMS coating had the lowest fouling release performance of about 0.2 MPa which had lowest surface energy and elastic modulus. As the elastic modulus of the PDMS-zPDEM-x coating increased, the removal strength of the coating ranged from 0.3 MPa to 0.5 MPa, which was slightly higher than that of the

PDMS coating. The PDMS-zPDEM-x coatings continued to exhibit low removal strength of pseudobarnacles, which indicated that the coatings with zwitterionic still had good fouling release ability [22].

In order to prove the stability of the coatings in seawater, the immersion weight loss experiment of the coating was carried out. Theoretically, PDEM(OH)₂ introduced into the reaction system as a chain extender could form a network structure under the action of the cross-linking agent so that it would not be released into the environment. As shown in Fig. 3f, the mass loss of PDMS-zPDEM-x was ranged from 1.06% to 1.38% and PDMS was about 0.4% after soaking in ASW for 21 days, because the unreacted substances and a small amount of solvent in the cross-linking network were precipitated under the immersion of ASW. Although the mass loss of PDMS-zPDEM-x was greater than that of PDMS, its weight loss was less than 2%, which proved that the coating was stable in ASW and could be used in marine [61].

3.5. Antifouling performance

Excellent antibacterial properties were necessary for antifouling coatings. So, the antimicrobial efficiency was evaluated by studying the number of colonies adhering to the coatings. As shown in Fig. 4a, the antimicrobial efficiency of PDMS-zPDEM-0 was 9.62%, because the hydrophobic and low surface energy properties of silicone weakened the interaction between the bacteria and the coating surface to reduce the adhesion. PDMS-zPDEM-x with different zwitterion contents had good antibacterial performance. Specifically, the antimicrobial efficiency of PDMS-zPDEM-5, PDMS-zPDEM-10 and PDMS-zPDEM-14 were 79.18%, 96.61% and 94.86%, indicating that zwitterions could improve the antimicrobial performance. In addition, biofilm was a part of marine fouling and the resistance of coatings to biofilm was conducted. Biofilm formation was induced by prolonging the incubation time of the *Pseudomonas* sp. The quality of biofilm was determined by absorbance at wavelength 589 nm after crystal violet staining, which was proportional to the quality of biofilm. It could be seen from Fig. 4b that the control sample was completely covered by biofilm with an OD of 2.66. The OD of PDMS-zPDEM-0 was 2.13, indicating that PDMS-zPDEM-0 had a slight resistance to biofilm formation and adhesion, it was consistent with the antimicrobial results. With the increase of zPDEM, the OD showed a decreasing trend of 2.06 for PDMS-zPDEM-5, 1.62 for PDMS-zPDEM-10 and 1.03 for PDMS-zPDEM-14, and the biofilm adhered to the coating surface gradually decrease. We also counted the area covered by the biofilm, and the statistics of the area covered by the biofilm remained consistent with the results of the OD. In general, the introduction of zPDEM resulted in the coating resistance to biofilm adhesion. Diatoms were typical of marine fouling organisms and *Pleurosigma* sp. was chosen to study the anti-diatom properties of the coatings. It could be seen from the Fig. 4c that lots of diatoms adhered to PDMS-zPDEM-0 (824 ± 213 cells/mm²) compared with the control group (638 ± 148 cells/mm²), because diatoms liked hydrophobic surfaces [62]. With the increase of zPDEM, the number of diatoms adhering to the coating showed a decreasing trend of 490 ± 167 cells/mm² for PDMS-zPDEM-5, 248 ± 105 cells/mm² for PDMS-zPDEM-10 and 107 ± 65 cells/mm² for PDMS-zPDEM-14. Due to the randomness of diatom culture, the error of experimental data was large, but this was similar to the report in the literature [63]. The result of anti-diatom experiment was consistent with the results of the antibacterial and the anti-biofilm tests, indicating that the introduction of zPDEM could improve the fouling resistance ability of coatings.

4. Conclusion

A new type of silicone low surface energy coating with zPDEM was designed. The coatings had good mechanical properties higher than 6.0 MPa and strong substrate adhesion up to 2.0 MPa. It was worth noting that the coating had low surface energy less than 30 mJ·m⁻² and exhibited good fouling release properties in the range of 0.3 MPa to 0.5

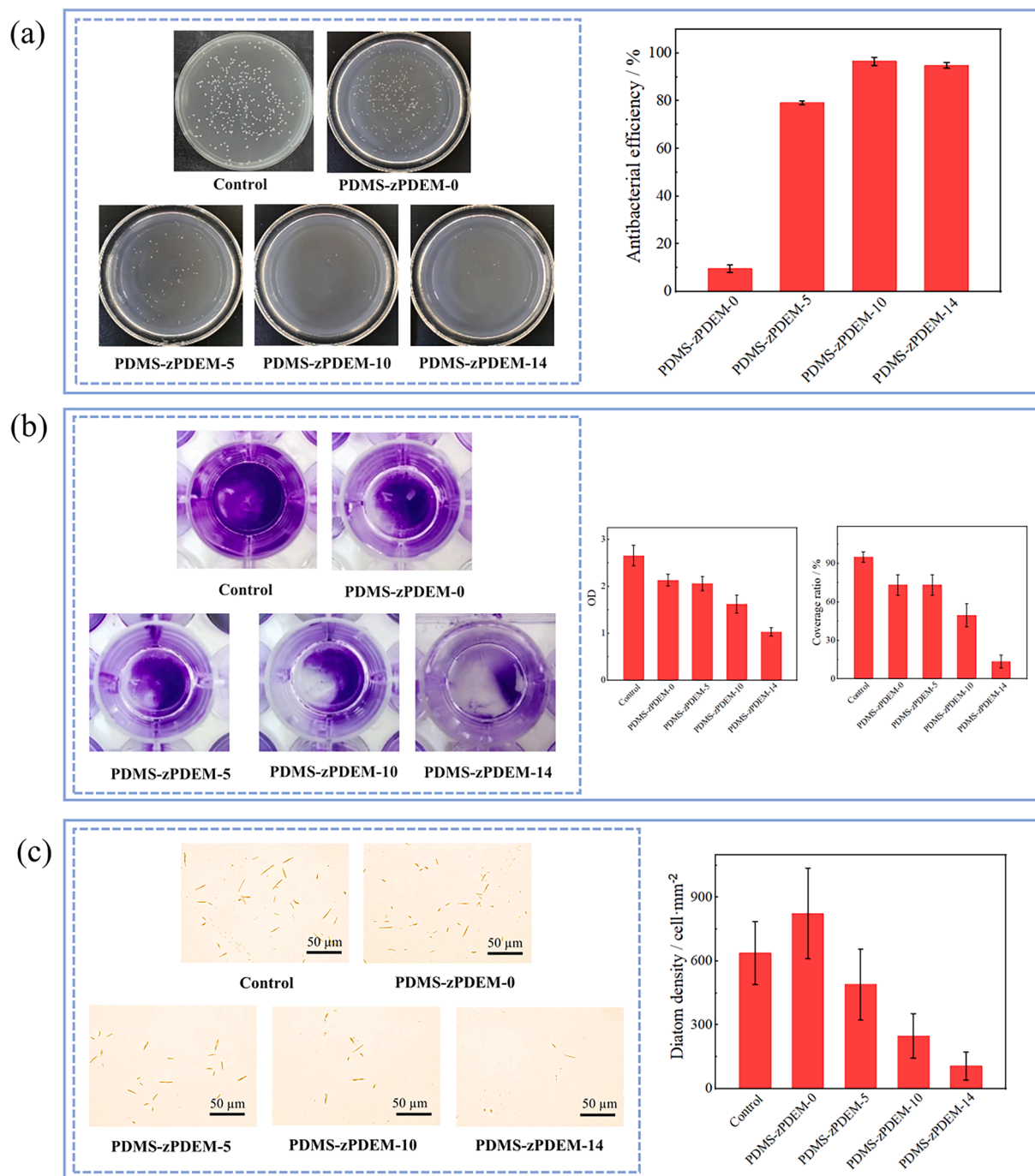


Fig. 4. Antibacterial image and antibacterial rates of PDMS-zPDEM-x for *Pseudomonas* sp. (a). The images of *Pseudomonas* sp. biofilm stained with crystal violet and the optical density (OD) of crystal violet residue in the biofilm at $\lambda = 589$ (b). The images and density of diatom (*Pleurosigma* sp.) adhered on PDMS-zPDEM-x (c). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

MPa. Meanwhile, the coating retained the original fouling release peculiarities of the silicone materials and could effectively prevent the adhesion of *Pseudomonas* sp. and the accumulation of biofilms and diatoms. This study provided a scenario for the design of low surface energy coatings and was expected to be applied in the field of marine antifouling coatings and biomaterial interfaces.

CRediT authorship contribution statement

Dejin Zhang: Conceptualization, Methodology, Data curation, Writing – original draft. **Susu Zhao:** Conceptualization, Methodology, Writing – review & editing. **Zhihao Rong:** Software, Investigation.

Kaiming Zhang: Writing – review & editing. **Chuanhui Gao:** Resources, Funding acquisition, Project administration. **Yumin Wu:** Visualization, Funding acquisition, Project administration. **Yuetao Liu:** Supervision, Project administration, Resources, Funding acquisition, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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